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Course Code : CSA 5111
Course Name : Cryptography and network security.

1) Plaintext : MEETATNOON

Caesar cipher with shift 7

$C = (P+7) \bmod 26$ $\quad P = (C-7) \bmod 26$.

Vigenere cipher key 'SUN'.

Plaintext : MEETATNOON

Key : 7

Caesar cipher Encryption $C = (P+7) \bmod 26$

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

Q	R	S	T	U	V	W	X	Y	Z
17	18	19	20	21	22	23	24	25	26

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Q	R	S	T	U	V	W	X	Y	Z
16	17	18	19	20	21	22	23	24	25

$C = (P+7) \bmod 26$

MEETATNOON

12 4 4 19 0 19 13 14 14 13

$C = (P+7) \bmod 26$

$= (12+7) \bmod 26$

$= 19 \bmod 26 = 19 = T$

$C = (4+7) \bmod 26 = 11 \bmod 26 = 11 = L$

$C = (4+7) \bmod 26 = 11 \bmod 26 = 11 = L$

$C = (19+7) \bmod 26 = 26 \bmod 26 = 0 = A$

$C = (0+7) \bmod 26 = 7 \bmod 26 = 7 = H$

$C = (19+7) \bmod 26 = 26 \bmod 26 = 0 = A$

$C = (13+7) \bmod 26 = 20 \bmod 26 = 20 = U$

$C = (14+7) \bmod 26 = 21 \bmod 26 = 21 = V$

$C = (14+7) \bmod 26 = 21 \bmod 26 = 21 = V$

$C = (13+7) \bmod 26 = 20 \bmod 26 = 20 = U$

MEETATNOON = TLLAHAUVVU

Decryption :

$P = (C-7) \bmod 26$

$P = (19-7) \bmod 26 = 12 \bmod 26 = 12 = M$

$P = (11-7) \bmod 26 = 4 \bmod 26 = 4 = E$

$P = (11-7) \bmod 26 = 4 \bmod 26 = 4 = E$

$P = (19-7) \bmod 26 = 19 \bmod 26 = 19 = T$

$P = (7-7) \bmod 26 = 0 = A$

$P = (0-7) \bmod 26 = 19 \bmod 26 = 19 = T$

$= (20-7) \bmod 26 = 13 \bmod 26 = 13 = N$

$= (21-7) \bmod 26 = 14 \bmod 26 = 14 = O$

$P = (21-7) \bmod 26 = 14 \bmod 26 = 14 = O$

$P = (20-7) \bmod 26 = 13 \bmod 26 = 13 = N$

re cipher

ntext :	12 M	4 E	4 E	19 T	0 A	19 T	13 N	14 O	14 O	13 N
Key :	S	U	N	S	U	N	S	U	N	S
	18	20	13	18	20	13	18	20	13	18

Encryption:

$C = (P+K) \bmod 26$

$C = (12+18) \bmod 26 = 30 \bmod 26 = 4 = E$

$C = (4+20) \bmod 26 = 24 \bmod 26 = 24 = Y$

$C = (4+13) \bmod 26 = 17 \bmod 26 = 17 = R$

$C = (19+18) \bmod 26 = 37 \bmod 26 = 11 = L$

$C = (0+20) \bmod 26 = 20 \bmod 26 = 20 = U$

$C = (19+13) \bmod 26 = 32 \bmod 26 = 6 = G$

$C = (13+18) \bmod 26 = 31 \bmod 26 = 5 = F$

$C = (14+20) \bmod 26 = 34 \bmod 26 = 8 = I$

$C = (14+13) \bmod 26 = 27 \bmod 26 = 1 = B$

$C = (13+18) \bmod 26 = 31 \bmod 26 = 5 = F$

$= EYRLUGFIBF$

Decryption;

$P = (C-K) \bmod 26$

$P = (4-18) \bmod 26 = 12 \bmod 26 = 12 = M$

$P = (24-20) \bmod 26 = 4 \bmod 26 = 4 = E$

$P = (17-13) \bmod 26 = 4 \bmod 26 = 4 = E$

$P = (11-18) \bmod 26 = 19 \bmod 26 = 19 = T$

$P = (20-20) \bmod 26 =$

$P = (6-13) \bmod 26 = 19 \bmod 26 = 19 = T$

$P = (5-18) \bmod 26 = 13 \bmod 26 = 13 = N$

$P = (8-20) \bmod 26 = 14 \bmod 26 = 14 = O$

$P = (1-13) \bmod 26 = 14 \bmod 26 = 14 = O$

$P = (5-18) \bmod 26 = 13 \bmod 26 = 13 = N$

MEETATNOON

2) Plaintext : MATH

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

$$K = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

Hill cipher = MA/TH

12 0/19 7

Encryption :

Encrypt MA $= \begin{bmatrix} 12 \\ 0 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$

$= (36, 60) \bmod 26$

$= 36 \bmod 26 = 10 = K$

$= 60 \bmod 26 = 8 = I$

Encrypt TH $= \begin{bmatrix} 19 \\ 7 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$

$= (71, 144) \bmod 26$

$(19, 14) = TO$

2 H = KITO

ryption

$$\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

$$\frac{1}{ad-bc} = \frac{1}{3(7)-2(5)} = \frac{1}{21-10} = \frac{1}{11} = 11^{-1} = 19$$

$$\frac{1}{19}\begin{bmatrix} 19\begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \end{bmatrix}$$

$$\begin{bmatrix} 133 & -36 \\ -95 & 57 \end{bmatrix} \bmod 26$$

$-36 \bmod 26 = 16$

$-95 \bmod 26 = 9$

$$\begin{bmatrix} 10 \\ 8 \end{bmatrix}\begin{bmatrix} 133 & 16 \\ 9 & 57 \end{bmatrix} \bmod 26$$

$= (1458 \quad 546) \bmod 26$

$= 12 \bmod 26 = 12 = M$

$= 0 \bmod 26 = 0 = A$

$$\begin{bmatrix} 19 \\ 14 \end{bmatrix}\begin{bmatrix} 133 & 16 \\ 9 & 57 \end{bmatrix} \bmod 26$$

$= (2751, 969) \bmod 26$

$= 19 \bmod 26 = 19 = T$

$= 7 \bmod 26 = 7 = H$

Decryption =

Key: $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$

MA | TH
12 0 | 19 7

$$\begin{bmatrix} 12 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \bmod 26 = (24, 48) \bmod 26$$

$$= 24 \bmod 26 = Y$$

$$48 \bmod 26 = W$$

$$\begin{bmatrix} 19 \\ 7 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \bmod 26 = (59, 111) \bmod 26$$

$$= (7, 7) = H, H$$

Decryption: $\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$

$$\frac{1}{ad-bc} = \frac{1}{2(5)-3(4)} = \frac{1}{10-12} = \frac{1}{-2} =$$

$$2^{-1} \begin{bmatrix} 5 & -3 \\ -4 & 2 \end{bmatrix}$$

The key $\begin{bmatrix} 2 & +3 \\ 4 & 5 \end{bmatrix}$ is unacceptable. It is an invalid key.

arison between AES and RSA

ture	AES	RSA
y usage	Same secret key for encryption & decryption	Encryption using public key & Decryption using private key
Performance	Very fast	slower
Data size	It is suitable for large files like 1GB	Not suitable for large files.
Key Distribution	Secret key must be shared securely.	Public key can be shared openly.

No. of keys required

200 clinics

$$\frac{200 \times 199}{2} = 19,900$$

Using RSA each clinic has one public and private key.

Total = 201 RSA key pairs.

Hybrid Encryption Workflow:

Hybrid encryption combines AES and RSA to provide Security and speed.

1) Sending clinic steps

i) Verify the server's digital certificate.

ii) Generate a random 256-bit.

iii) Encrypt the MRI file using AES-256.

iv) Encrypt the DEK using RSA.

2) Receiving Server steps
   i) Receive the encrypted package.
   ii) Use RSA key to recover DEK.
   iii) Use AES-GCM to decrypt MRI file.
   iv) Verify the authentication tag if it is valid otherwise reject.

3) Secure key storage
   i) Store keys and keep private keys non-exportable.
   ii) Allow access only to authorized.
   iii) Use MFA for extra security.
   iv) Maintain audit logs.

4) Key Rotation
   i) Generate a separate DEK and change RSA keys.
   ii) Re-encrypt using new RSA public key.
   iii) Immediately replace keys if they are suspected.

The secure key should be handled (HSM/KMS) non exportable keys, role based access, MFA, audit logging backup and recovery.

The key rotation should be handled (per-file DEKs, periodic rotation of the key-encryption key-encryption key, re-wrapping, retirement of the old keys, and emergency rotation after the suspected compromise.

, $q = 19$

$= 7$, Message $M = \xi = M^e \bmod n$ $\quad M = C^d \bmod n$

$1 = n \quad \gcd(e, \phi(n)) = 1$

$n = p \times q$

$= 11 \times 19 = 209 \qquad n = 209$

$\phi_n = (p-1)(q-1)$

$= (11-1)(19-1)$

$= 10 \times 18 = 180 \qquad \phi_n = 180$

$\gcd(7, 180) = 1$

$1 < e < \phi_n$

$1 < 7 < 180$

$e \cdot d = n$

$7d = 1 \bmod 180$

$d = 103$

Encryption $\quad C = M^e \bmod n$

$= 5^7 \bmod 209$

$C = 168$

Decryption $\quad M = C^d \bmod n$

$= 168^{103} \bmod 209$

$M = 5$

5) Plaintext: PAYNOW

Key : 7

Caesar cipher

P	A	Y	N	O	W
15	0	24	13	14	22

Encryption $C = (P+7) \bmod 26$

$C = (15+7) \bmod 26 = 22 \bmod 26 = 22 = W$

$C = (0+7) \bmod 26 = 7 \bmod 26 = 7 = H$

$C = (24+7) \bmod 26 = 31 \bmod 26 = 5 = F$

$C = (13+7) \bmod 26 = 20 \bmod 26 = 20 = U$

$C = (14+7) \bmod 26 = 21 \bmod 26 = 21 = V$

$C = (22+7) \bmod 26 = 29 \bmod 26 = 3 = D$

$= WHFUVD$

Decryption $P = (C-7) \bmod 26$

$P = (22-7) \bmod 26 = 15 \bmod 26 = 15 = P$

$P = (7-7) \bmod 26 = 0 = A$

$P = (5-7) \bmod 26 = 24 \bmod 26 = Y$

$P = (20-7) \bmod 26 = 13 \bmod 26 = 13 = N$

$P = (14-7) \bmod 26 = 14 \bmod 26 = 14 = O$

$P = (3-7) \bmod 26 = 22 \bmod 26 = 22 = W$

$= PAYNOW$

~~MONARCHY~~ PAYNOW

: MONARCHY

M	O	N	A	R	C	H	Y
B	C	D	E	F	G	I	

M	O	N	A	R
C	H	Y	B	C
D	E	F	G	I
J	K	L	P	Q
S	T	U	V	

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Encryption : PA/YN/OW

PA = SO

YN = GY

OW = NV

Decryption SO = PA

GY = YN

NV = OW

Vigenere

12 4 4 19 0 19 13 14 14 13

Plaintext: M E E T A T N O O N

Key : S U N S U N S U N S

18 20 13 18 20 13 18 20 13 18

Encryption: $C = (P+K) \bmod 26$

$C = (12+18) \bmod 26 = 30 \bmod 26 = 4 = E$

$C = (4+20) \bmod 26 = 24 \bmod 26 = 24 = Y$

$C = (4+13) \bmod 26 = 17 \bmod 26 = 17 = R$

$C = (19+18) \bmod 26 = 37 \bmod 26 = 20 = U$

$C = (0+20) \bmod 26 = 20 \bmod 26 = 20 = U$

$C = (19+13) \bmod 26 = 32 \bmod 26 = 6 = G$

$C = (13+18) \bmod 26 = 31 \bmod 26 = 5 = F$

$C = (14+20) \bmod 26 = 34 \bmod 26 = 8 = I$

$C = (14+13) \bmod 26 = 27 \bmod 26 = 1 = B$

$C = (13+18) \bmod 26 = 31 \bmod 26 = 5 = F$

EYRLUGFIBF

Decryption:

$P = (C-K) \bmod 26$

$P = (4-18) \bmod 26 = 12 \bmod 26 = 12 = M$

$P = (24-20) \bmod 26 = 4 \bmod 26 = 4 = E$

$P = (17-13) \bmod 26 = 4 \bmod 26 = 4 = E$

$P = (11-18) \bmod 26 = 19 \bmod 26 = 19 = T$

$P = (20-20) \bmod 26 = 0 = A$

$P = (6-13) \bmod 26 = 19 \bmod 26 = 19 = T$

$P = (5-18) \bmod 26 = 13 \bmod 26 = 13 = N$

$P = (8-20) \bmod 26 = 14 \bmod 26 = 14 = O$

$P = (1-13) \bmod 26 = 14 \bmod 26 = 14 = O$

$P = (5-18) \bmod 26 = 13 \bmod 26 = 13 = N$